

Tom Huth

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PROFILE

Full-stack software developer with a BSc in Computing Science from the University of Glasgow and a strong track record of independently designing, building, and deploying complex multi-role web applications backed by relational databases. Experienced in the full development lifecycle from requirements analysis and data modelling through to testing, documentation, and deployment, with particular expertise in Node.js, PostgreSQL, RESTful API design, and modern front-end frameworks. Brings a rigorous approach to security, validation, and data integrity, and is comfortable producing technical documentation and maintaining high standards of code quality across multiple concurrent projects.

TECHNICAL SKILLS

Languages & Frameworks

JavaScript / TypeScript, Node.js, Python, Java, SQL, HTML/CSS, React (Vite), Angular 17, Django, Express, Spring

Databases

PostgreSQL, MySQL, MongoDB, Prisma ORM

Relational data modelling, complex queries, schema design, referential integrity

Development Practices

RESTful API design, JWT authentication, input validation (Zod), rate limiting, automated testing (Jest / Supertest / Vitest), Git, Docker, Linux/Unix, deployment

Documentation & Research

Technical specifications, test plans, experimental reporting, academic and professional written communication

SOFTWARE DEVELOPMENT PROJECTS

Etradie – Full-Stack Marketplace Platform | Personal Project (Founder)

2025–2026

End-to-end marketplace connecting homeowners with tradespeople, built and deployed independently. Demonstrates full ownership of the software development lifecycle from requirements through to production.

Stack: React (Vite), Node.js, Express, PostgreSQL, Prisma ORM, Socket.IO, JWT

GitHub: github.com/TomHuth9/Etradie

- Designed and implemented a relational PostgreSQL schema supporting complex multi-role user data (homeowners, tradespeople), job records, messaging threads, reviews, and notifications, with full referential integrity.
- Built a secure RESTful API with JWT authentication, Zod input validation, structured route architecture, and rate limiting to protect data integrity and prevent abuse.
- Implemented real-time job broadcast to nearby tradespeople (distance-based filtering), per-job messaging via Socket.IO, and in-app notification system for key workflow events.
- Wrote automated test suites using Jest/Supertest (back end) and Vitest (front end), covering critical API routes and user flows.
- Produced technical documentation covering system architecture, API specifications, and deployment configuration.

The Farm Gate – Full-Stack Delivery Platform | Personal Project (Founder)

2026

Farm-to-door e-commerce platform with a REST API back end and role-based producer and consumer accounts.

Stack: MEAN (MongoDB, Express, Angular 17, Node.js), Angular Material

GitHub: github.com/TomHuth9/TheFarmGate

- Designed role-based account system (customers and farm producers) with dedicated producer dashboard for managing product listings, pricing, and images.
- Implemented basket management, order placement with postcode-based delivery centre assignment, and server-side price verification to maintain data integrity.
- Secured with JWT authentication, input validation, and rate limiting; comprehensive test coverage across the application.

Dissertation – CNN Generalisation on Visual Object Counting | University of Glasgow 2025–2026

Grade A3 (1st) research project requiring rigorous experimental design, data pipeline development, and detailed documentation, skills directly transferable to a validated, regulated software environment.

Stack: Python, PyTorch, NumPy, Matplotlib

GitHub: github.com/TomHuth9/Counting-Beyond-the-Training-Range-Evaluating-CNN-Generalisation-on-Visual-Object-Counting

- Designed a controlled experimental framework across 7 CNN architectures and 40+ model runs; built a synthetic dataset of 21,000 labelled images using a reproducible pipeline.
- Implemented custom evaluation metrics, experiment tracking, and ablation studies, producing detailed technical reporting throughout.
- Demonstrated ability to manage a complex, multi-component technical project independently, document methodology rigorously, and deliver to a strict deadline.

EDUCATION

BSc Computing Science (Faster Route) with Data Science Strand | University of Glasgow 2023 – 2026

Achieved 2:1 · Relevant modules: Web Application Development, Object-Oriented Software Engineering, Data Fundamentals, AI, Machine Learning, Human-Centred Systems Design, Robotics Foundations, Programming Languages

A-Levels | Our Lady and St Patrick's College, Belfast

2016 – 2023

Computer Science – A · Mathematics – A · Geography – A · Business Studies (AS) – B

WORK EXPERIENCE

Reach Out IT Student Ambassador | University of Glasgow

October 2024 – Present

- Provide first-line technical support to staff and students; troubleshoot software, hardware, and network issues across Windows, Office 365, and university-managed systems.
- Use Ivanti helpdesk platform for incident management; contribute to the team Knowledge Base with documented resolutions.
- Demonstrates ability to diagnose technical problems under pressure, communicate solutions clearly to non-technical users, and work within a structured institutional IT environment.

Senior General Counsellor | Camp Canada

Jun–Aug 2024 · Jun–Aug 2025

- Managed staff and campers in a high-responsibility environment; mentored junior colleagues and led project planning and conflict resolution.

Maintenance Technician | Smylie Sectional Buildings

Sep 2021 – Sep 2023

- Programmed and maintained automated control systems; carried out general hardware maintenance.

LEADERSHIP & ADDITIONAL

Volunteer Senior Youth Leader – Koinonia Ireland (2020–2023): Coordinated logistics and team management for large-scale international events; progressed from participant to senior leader.

TBUC Ambassador Mentor (2021–2023): Organised and hosted community events for up to 500 attendees; holds OCN Level 2 Award in Uniting Communities through Leadership.

Further: Duke of Edinburgh Silver Award · Kainos virtual work experience (web development, cybersecurity with OWASP Juice Shop) · NHS Administrator · Lifeguard